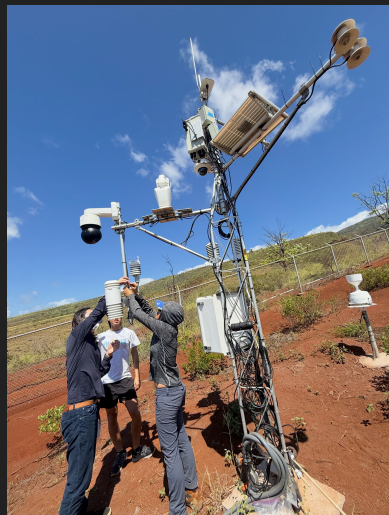


Hawaii Grande Update

Chris Lee, Jason Leigh, Ryan Theriot,
Anke Kuegler, Han Tseng, Chris Shuler, Tom Giambelluca



Overall Goals

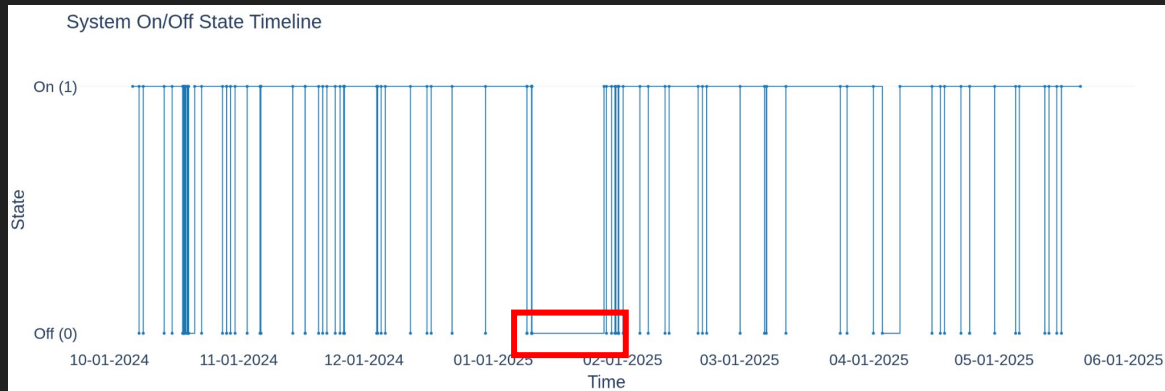
- Evaluate the efficacy and reliability of the Sage Sensor in Hawaii (primarily in Lahaina)
- To develop and evaluate AI-based disaster (fire/flood) detection capabilities
- Plan future Grande sites

Availability

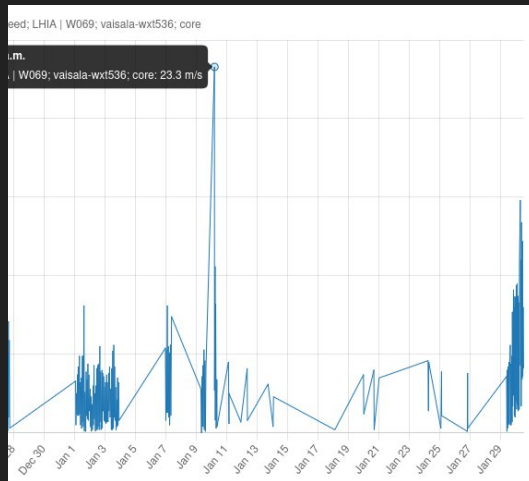
- Motivation:
 - Will the node be available when I need to access real time data or want to ssh into it?
- Availability:
 - How often is the node accessible to the users when they need it?

Availability

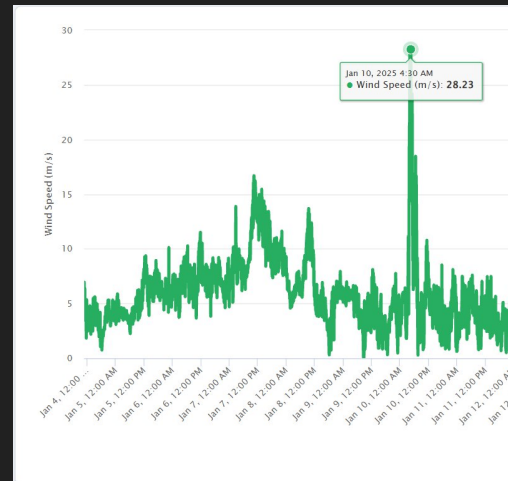
- Uptime 89.53%
- Total Downtime 23.78 days out of 227.09 days
- Starlink dish being dislodged due to high winds between January 10, 2025 - January 27, 2025.
 - High winds (23.3m/s - 51MPH) are believed to be the cause
 - Wind speeds cross referenced with mesonet (28.23m/s)



SAGE



Mesonet

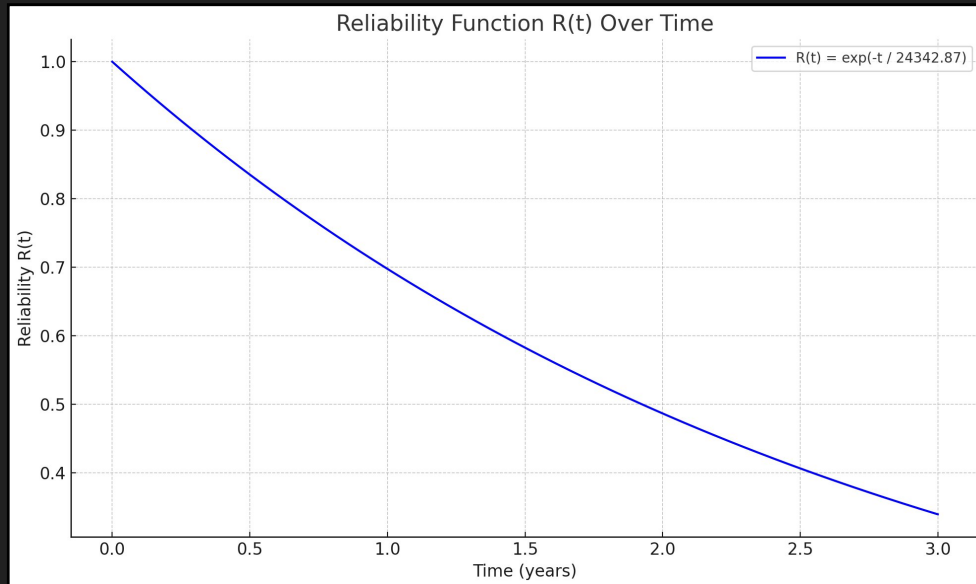


Reliability

- Motivation:
 - How often do we expect to have the system running without needing to send a technician to repair it.
- Reliability:
 - What's the probability that the node will not have component failures?

Mean Time Before Failure

- Analytical Model of Critical Computing Components
 - Xavier NX (CPU/GPU), Switches, USB, SSD, Modem
 - Components critical to data transmission from the instrumentation to Argonne's Servers
- Assumptions
 - Series (No Redundant Components)
 - Constant-Rate Model: each component has constant chance of failure over time, ignoring early failures and late stage wearout.
- MTBF **Analytical**:
 - ~2.78 years
 - Based on vendor published data sheet
- MTBF **Empirical** (under Hawaii conditions):
 - ~2.30 years
 - Based on Hawaii Nodes **W069** and **W097**



$$MTBF_{computeSys} = 1/\lambda_{computeSys} = (1/\sum_i (1/MTBF_i))$$

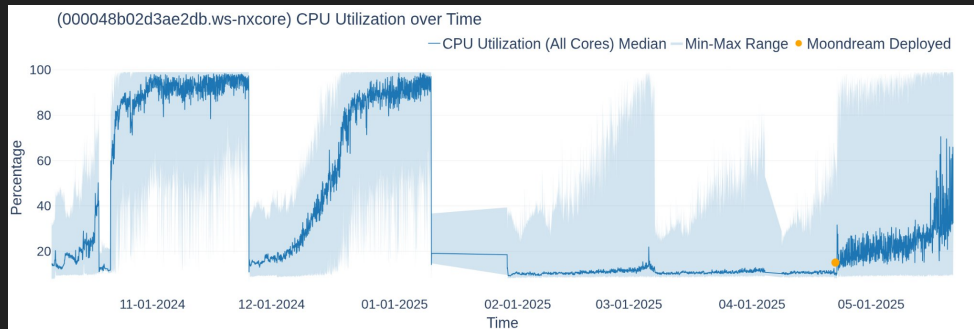
Metrics

- Motivation:
 - Any anomalies in the data, how well is the system performing?
 - Operating temperature, CPU, Storage, GPU under stress.
 - Network utilization

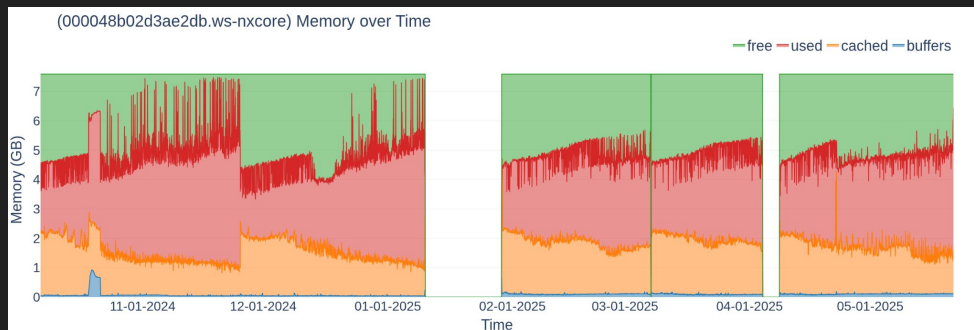
CPU1 (Xavier NX)

- CPU thermal highs
 - 61°C
- Average difference of ambient vs CPU-therm
 - 22.9°C
 - 10/6/2024 to 5/21/2025
- Max Safety Limit
 - 90.5°C
- Running AI Models such as Moondream on CPU1 (Describe scene) and Yolo (Fire Detection) on GPU

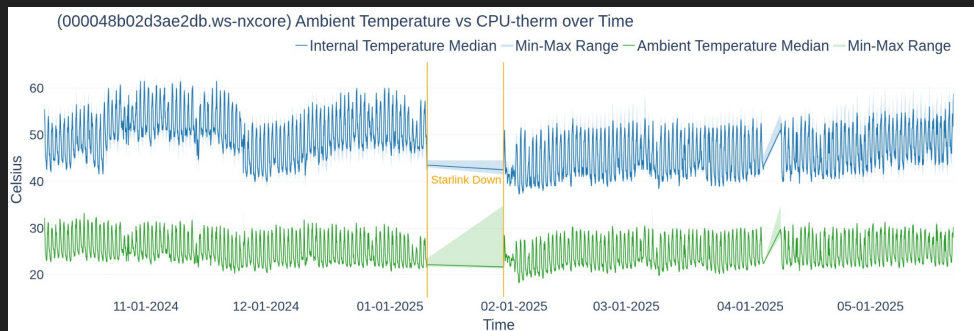
CPU



Mem



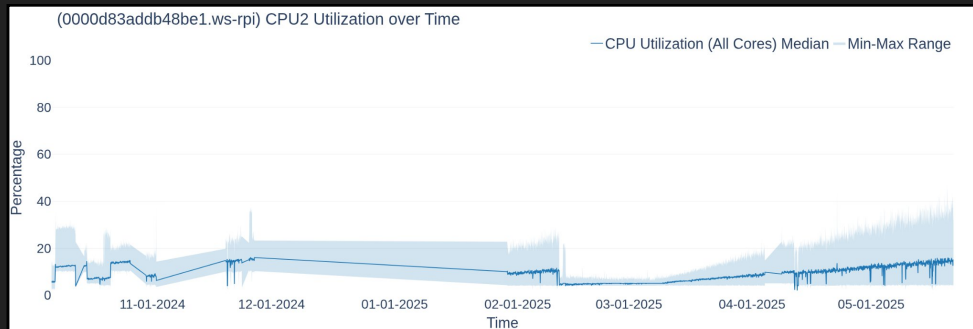
Temp



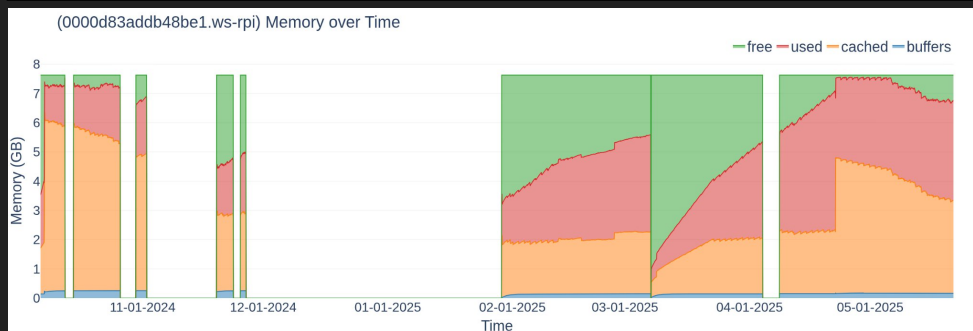
CPU2 (Raspberry PI)

- CPU thermal highs
 - 75°C to 80°C
- Max Safety Limit
 - 85°C
- Memory utilization high, but cpu seems underutilized
 - High Memory usage, maybe look into higher memory boards
 - We don't have ssh access to CPU2 to diagnosis the cause

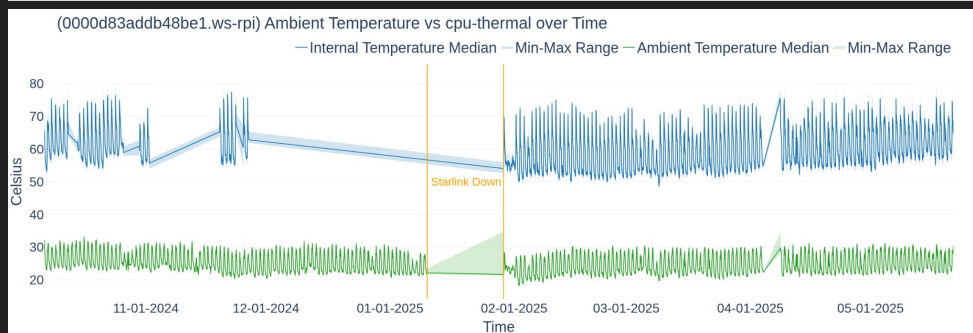
CPU



Mem



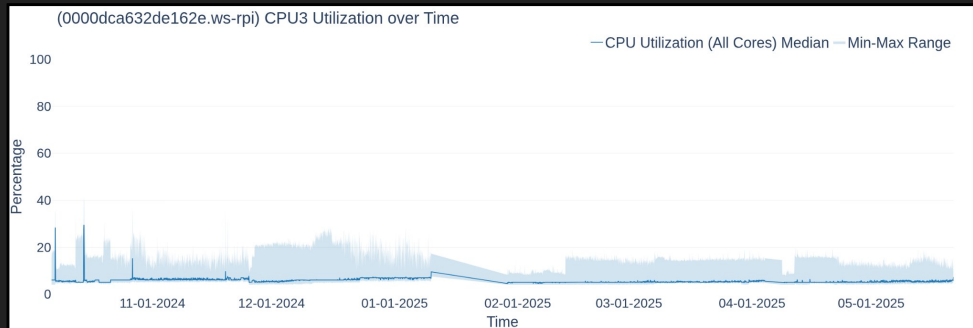
Temp



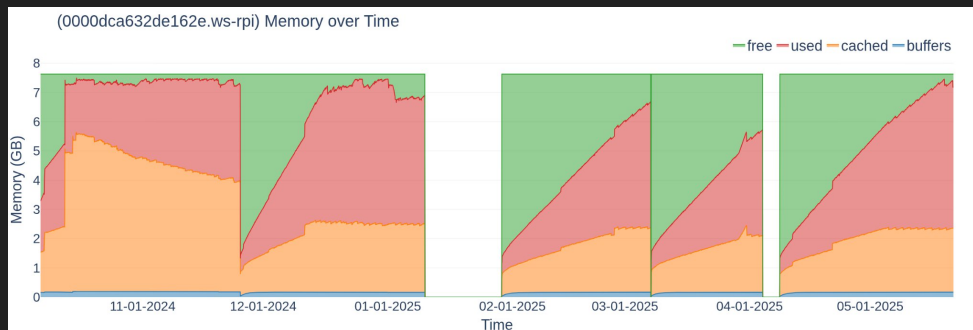
CPU3 (Raspberry PI)

- CPU thermal highs
 - 64°C
- Max Safety Limit
 - 85°C
- Memory utilization high, but cpu seems underutilized
 - Same observations as CPU2

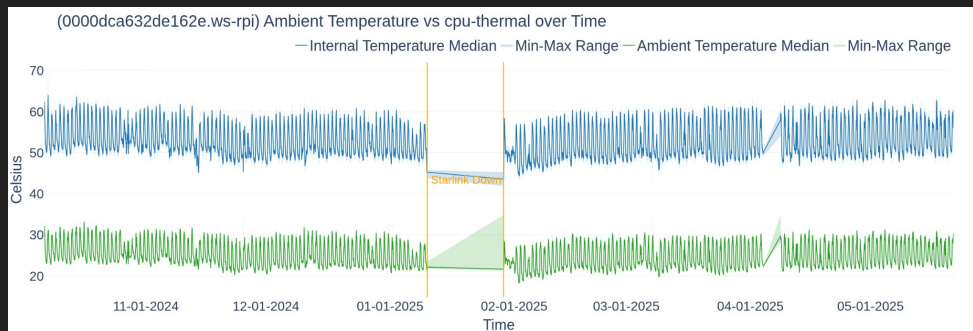
CPU



Mem

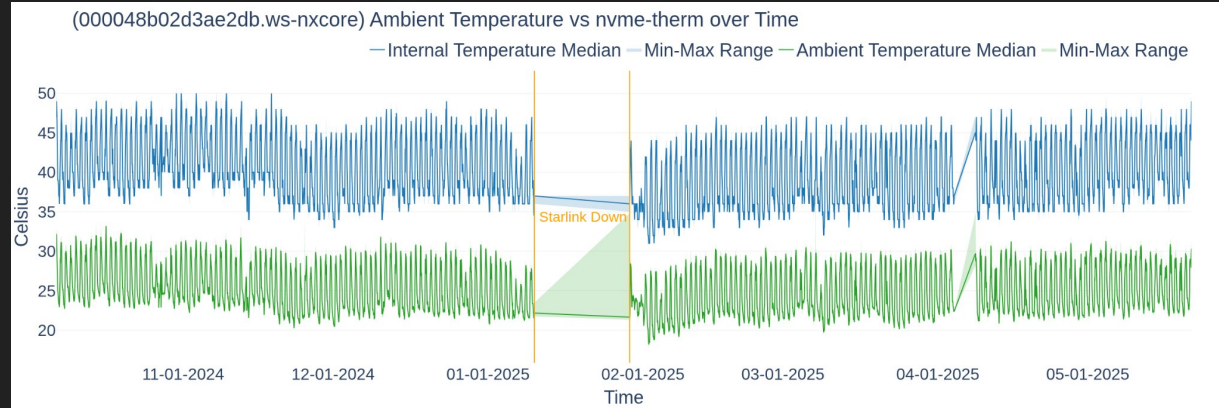


Temp



Storage Temperature

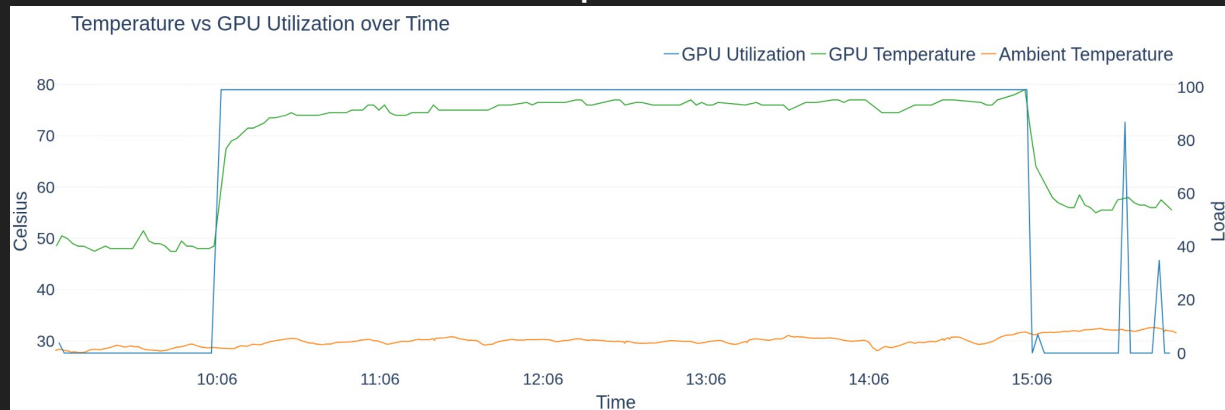
- NVME SSD thermals fall well within operating temperature (0°C to 70°C)
- Typically highs of 50°C and lows of 35°C
- Periodic fluctuations that are influenced by the ambient temperature.
- Higher temperatures during the day, lower during the night



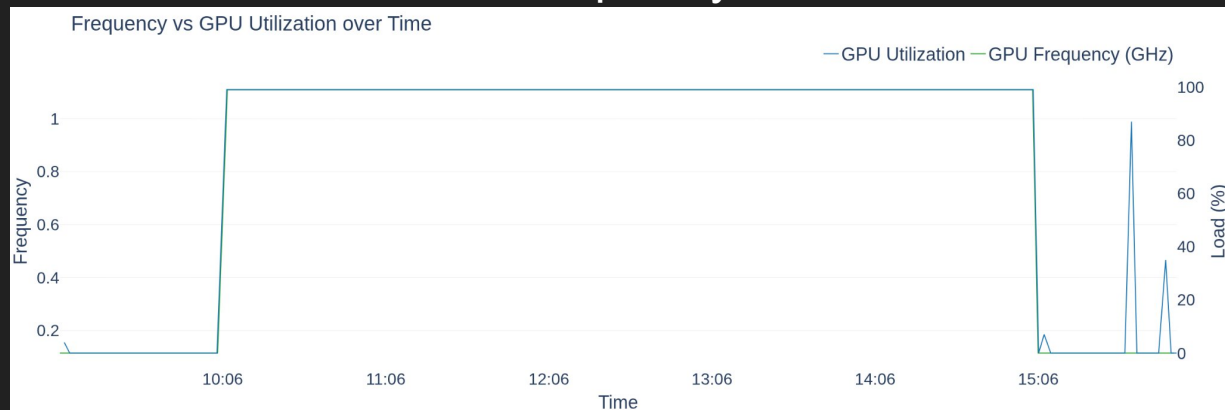
GPU Stress Test

- Tested between 10a-3p
 - Hottest ambient temp in Lahaina
- GPU temp
 - Average **77°C**
 - Peak **80°C** (at end)
- No dips in frequency observed
 - assumed no thermal throttling
- Max Safety Limit
 - **91.5°C**

Temperature

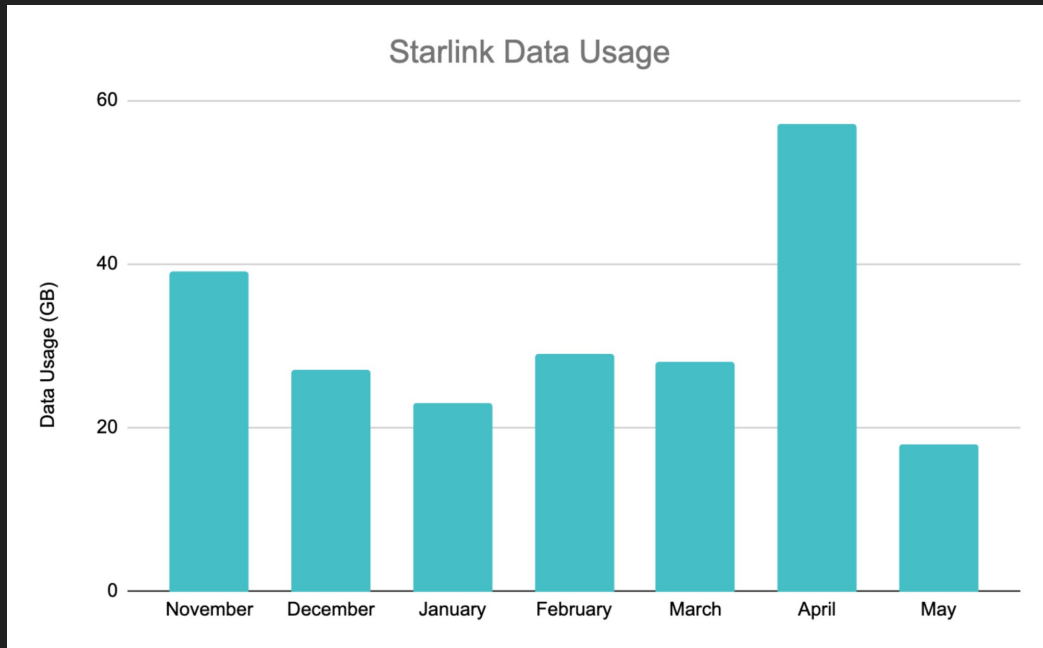
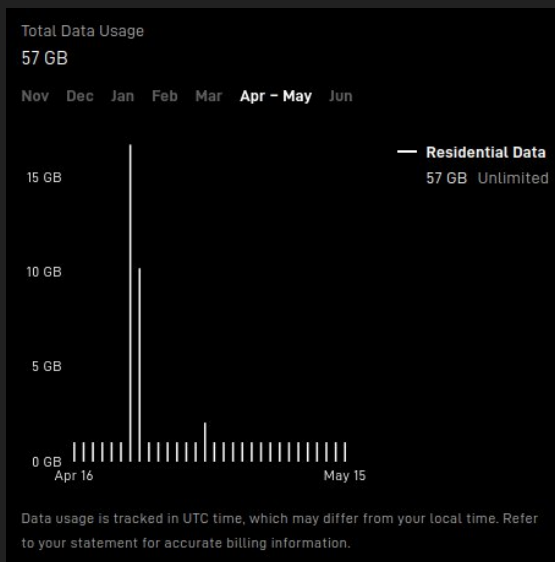


Frequency



Network Usage

- Disregarding spikes, on average, approximately **1GB per day**
- Spikes Because
 - Pulling docker containers/ kube pods
- Total: ~221GB (Nov to May)



Network Usage

TCP throughput appears low, maybe alternative congestion control should be considered for Starlink

Lahaina Node: (Starlink)

- Latency ranged between 134ms and 409ms. Round trip time mean deviation/jitter was 37ms.
- UDP achieved 17.5 Mb/s up and 20.1 Mb/s down
- TCP achieved 1.39 Mb/s up and 1.5 Mb/s down

Volcano node: (Cellular LTE)

- Latency ranged between 125ms and 132ms with jitter at 4.3ms.
- UDP bandwidth achieved 841 Kb/s up and 31.1 Mb/s down
- TCP achieved 1.74Mb/s up and 1.82Mb/s down

The use of Low Earth Orbit (LEO) satellites in systems like Starlink introduces frequent handoffs and dynamic routing, which significantly impact TCP performance (Handley, 2018). Satellite handoffs disrupt path stability, causing IP routing changes, brief disconnects, and packet loss or latency spikes. TCP misinterprets these losses as congestion, reducing its sending rate and leading to throughput drops that take time to recover.

LEO networks also suffer from variable latency. Since TCP relies on round-trip time (RTT) estimates to adjust data flow, unstable latency leads to under- or over-utilization of bandwidth. Frequent path changes mid-connection break session-level optimizations like TCP Fast Open, window scaling, and selective acknowledgment (SACK). Additionally, retransmissions can become mistimed due to volatile RTT, causing premature or delayed recovery and further inefficiency.

These challenges can be mitigated with TCP BBR, a modern congestion control algorithm that estimates bandwidth and RTT independently of packet loss. BBR maintains high throughput and low latency even in variable or lossy networks such as satellite links (Cardwell et al., 2017).

Handley, M. (2018). *Delay is Not an Option: Low Latency Routing in Space*. HotNets '18, pp. 85–91. <https://doi.org/10.1145/3286062.3286075>

Cardwell, N., Cheng, Y., Gunn, C. S., Yeganeh, S. H., & Jacobson, V. (2017). *BBR: Congestion-Based Congestion Control*. Communications of the ACM, 60(2), 58–66. <https://doi.org/10.1145/3009824>

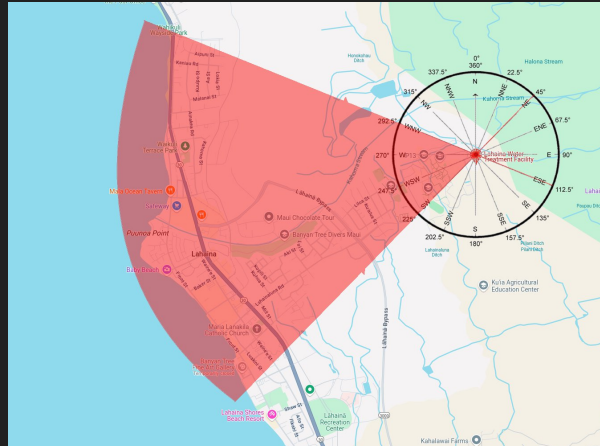
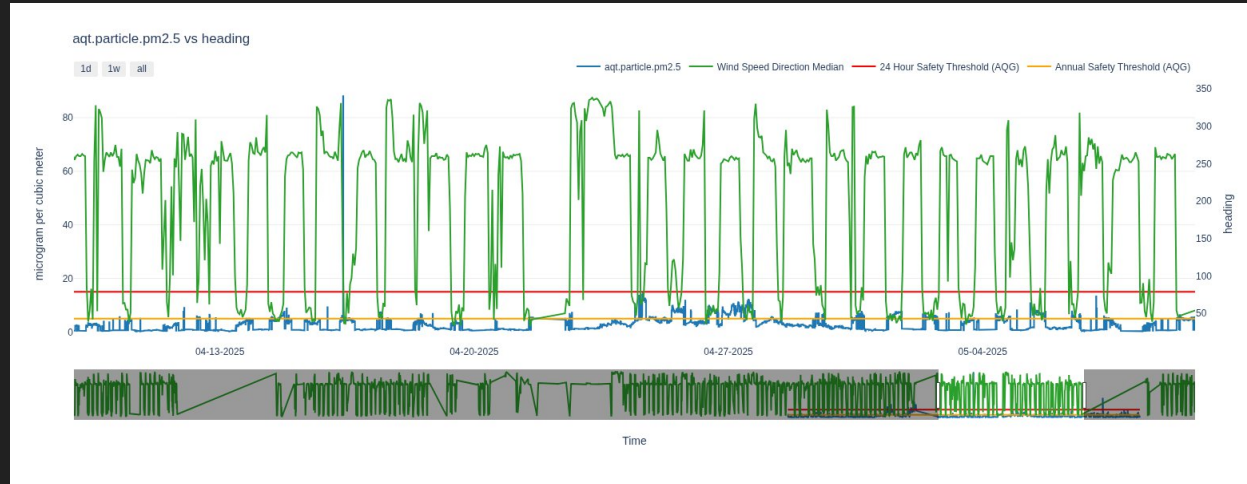
Huston, G., *Measuring Starlink Protocol Performance*: <https://www.youtube.com/watch?v=P7bk84H9bVs>

Pollutants

- Motivation:
 - It is safe out there? 🤖
- Pollutants:
 - Data gathered with Vaisala AQT530 (air quality) and safety compared to World Health Organization guidelines.

PM2.5

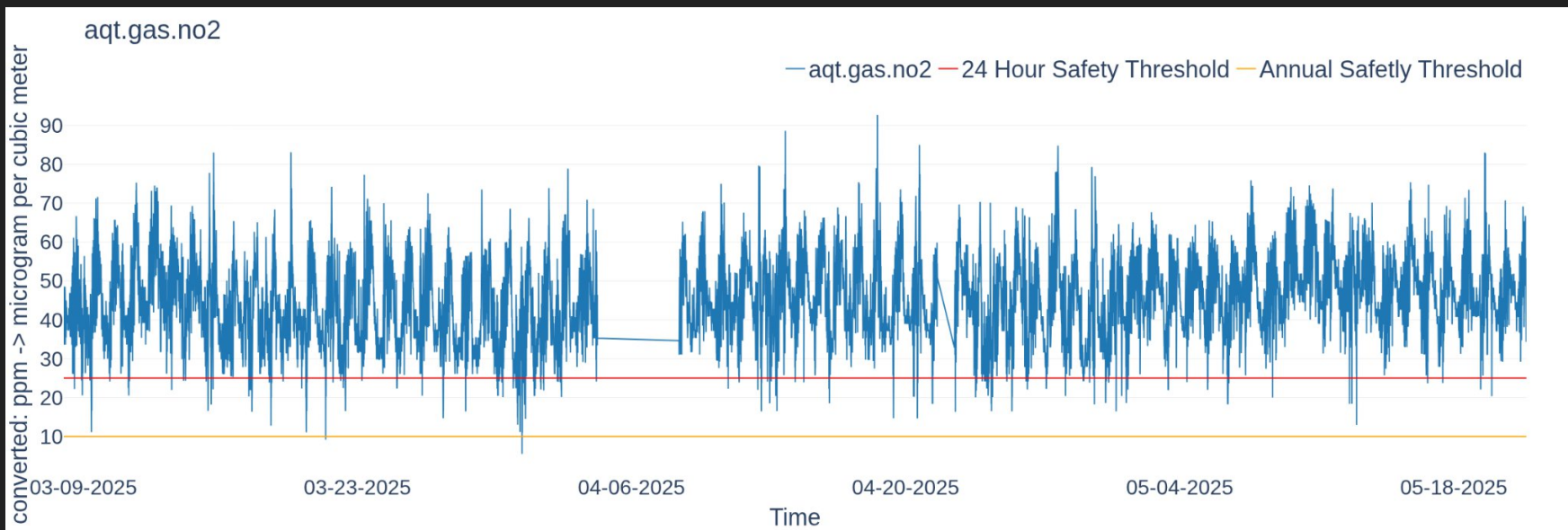
- **PM2.5** : Particulate matter with a diameter of 2.5 micrometers or less
 - Example source: vehicle exhaust, power plants
- Safety limits are based on **Air Quality Guide** set by **World Health Organization**
- Spikes in PM2.5 observed, theorized to be caused by **Lahaina construction or traffic during the day**
 - Based on visual correlations between wind direction and PM2.5 elevations
 - Sea breeze during the day and land breeze at night



- Cone Formed by 45° to 112.5° Heading

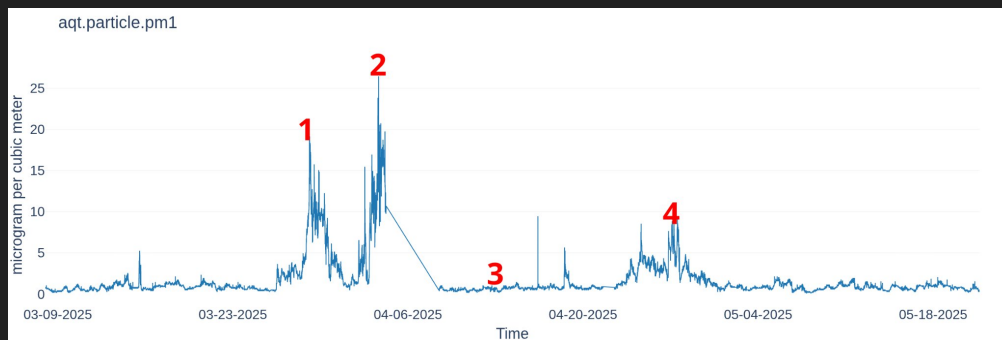
NO₂

- High NO₂ readings observed (compared to most stringent WHO recommendations)
- Want to verify with secondary source.



PM1

- **PM1** : Particulate matter with a diameter of 1 micrometers or less
 - Example source: vehicle exhaust, combustion, fires
- PM1 readings correlate with visual obfuscation of Lanai



AI/ML

- Motivation:
 - To develop detections for a wide range of spontaneous disaster-like events (fires, floods, eruptions, etc.)

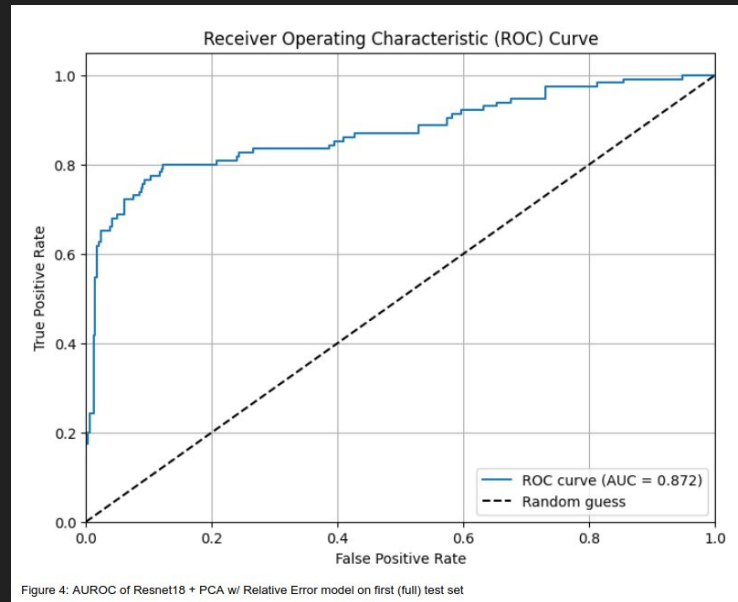
Fire Detection

- Method:
 - YOLO v7 on GPU (Fine-tuned on D-Fire training dataset)
- Evaluation
 - D-Fire Dataset (testing set)
 - Accuracy: 72.0% (smoke + fire)
 - Precision: 74.0%
 - Recall: 73.4%
 - F1 Score: 74%
- Performance Highlights:
 - The model shows high effectiveness in detecting smoke but less accuracy in identifying fire
- Challenges:
 - Needs to be specifically trained to detect certain classes (i.e., fire, smoke, floods)
- App: <https://portal.sagecontinuum.org/apps/app/giorgio808/yolov7-fire>



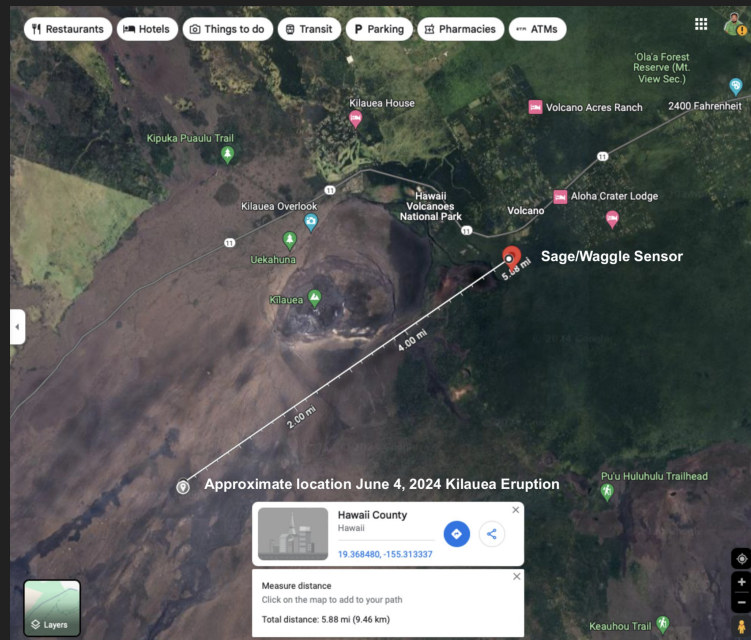
Anomaly Detection

- Motivation:
 - No need to train for specific disasters (fire, floods, smoke, etc.)
- Method:
 - Resnet18 (as feature map) into PCA (with relative reconstruction)
- Benefits:
 - Small training set required (~8 images)
 - Resnet helps reduce noise from tree swaying
- Evaluation:
 - 0.872 AUROC on full test set (has **minor anomalies and extreme anomalies**)
 - **67.8% anomaly accuracy** (class recall) on full test set
 - 82.7% anomaly accuracy (class recall) on extreme anomalies test set
- Challenges:
 - Reliability determined by user provided dataset and user adjustment of threshold
 - Not sensitive enough to detect smaller anomalies
 - Need to “re-train” the PCA if the PTZ moves, only favorable for static scenes



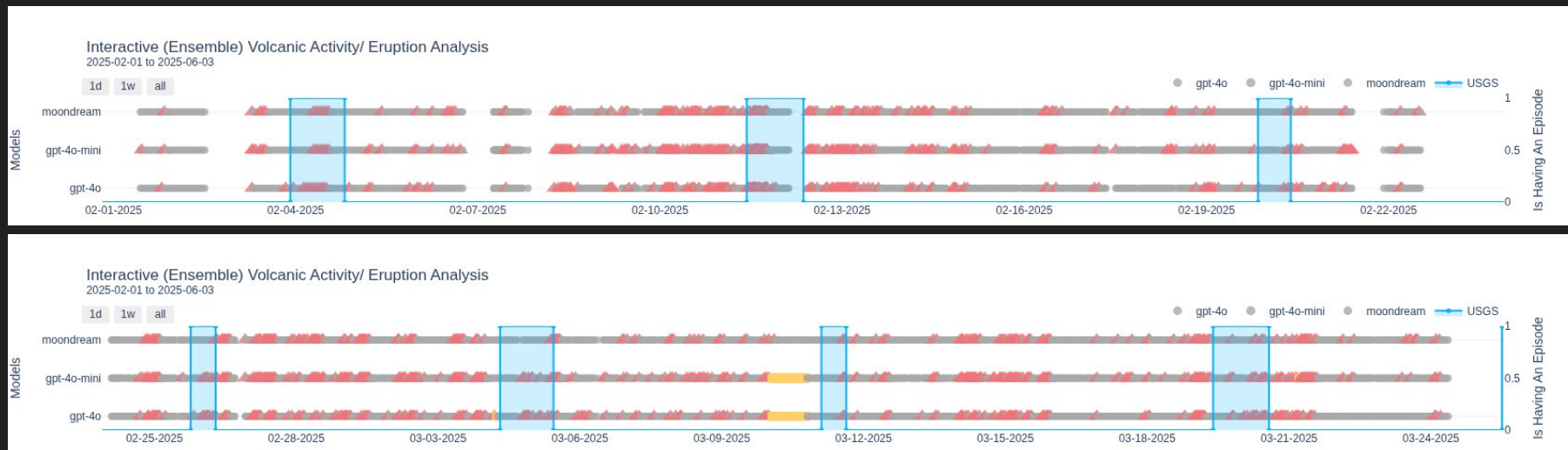
Vision Language Models

- Motivation:
 - Off the shelf VLMs come pre-trained and can be used on dynamically changing scenes.
- VLM off the shelf volcanic eruption detection (moondream2, gpt-4o-mini, gpt-4o)
- Prompt: Is there a volcanic eruption in this image? YES OR NO ANSWER
- VLM appears to be able to detect eruption when it sees lava
- VLM may detect an eruption if it sees smoke (not clear how it decides if it is an eruption)
- We can detect the presence of an eruption, but don't currently detect the end of the event. Whereas USGS only report eruptions for discrete events (episodes)
- We need our camera to be able to zoom in for a closer look during events
- More work needs to be done to characterize accuracy of detection, once the above issues are resolved



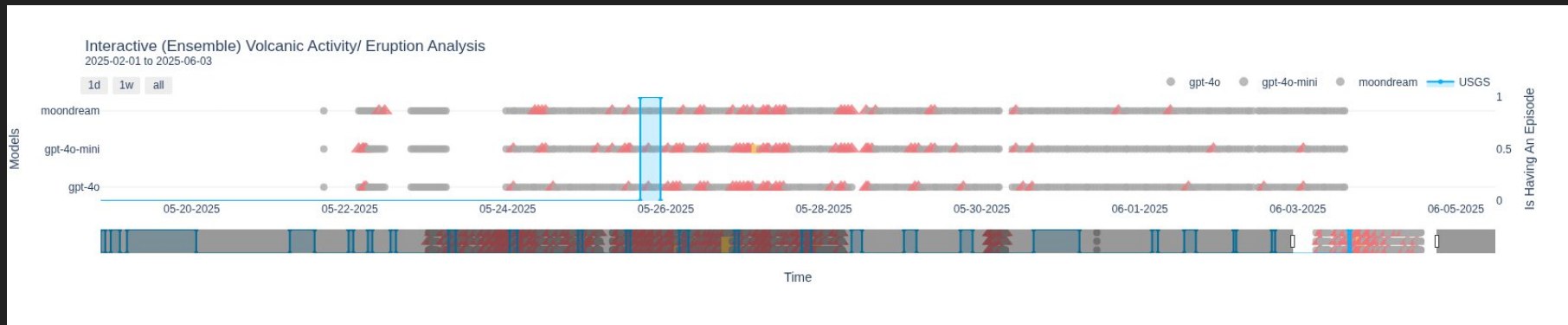
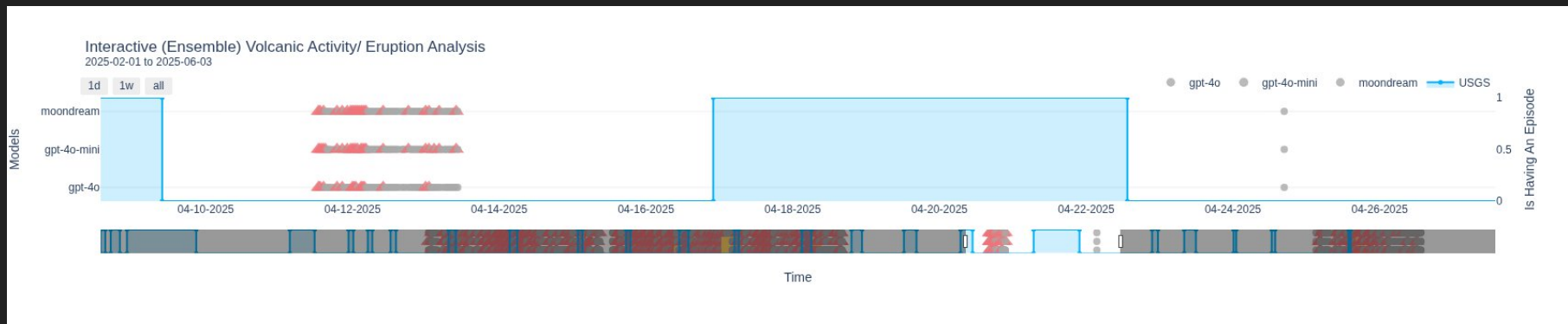


Visualization of Eruption Events



Blue box = discrete volcanic eruptions confirmed by USGS, Red = VLM detected eruption, Grey = VLM detected no eruption

(need a better way to visualize this)

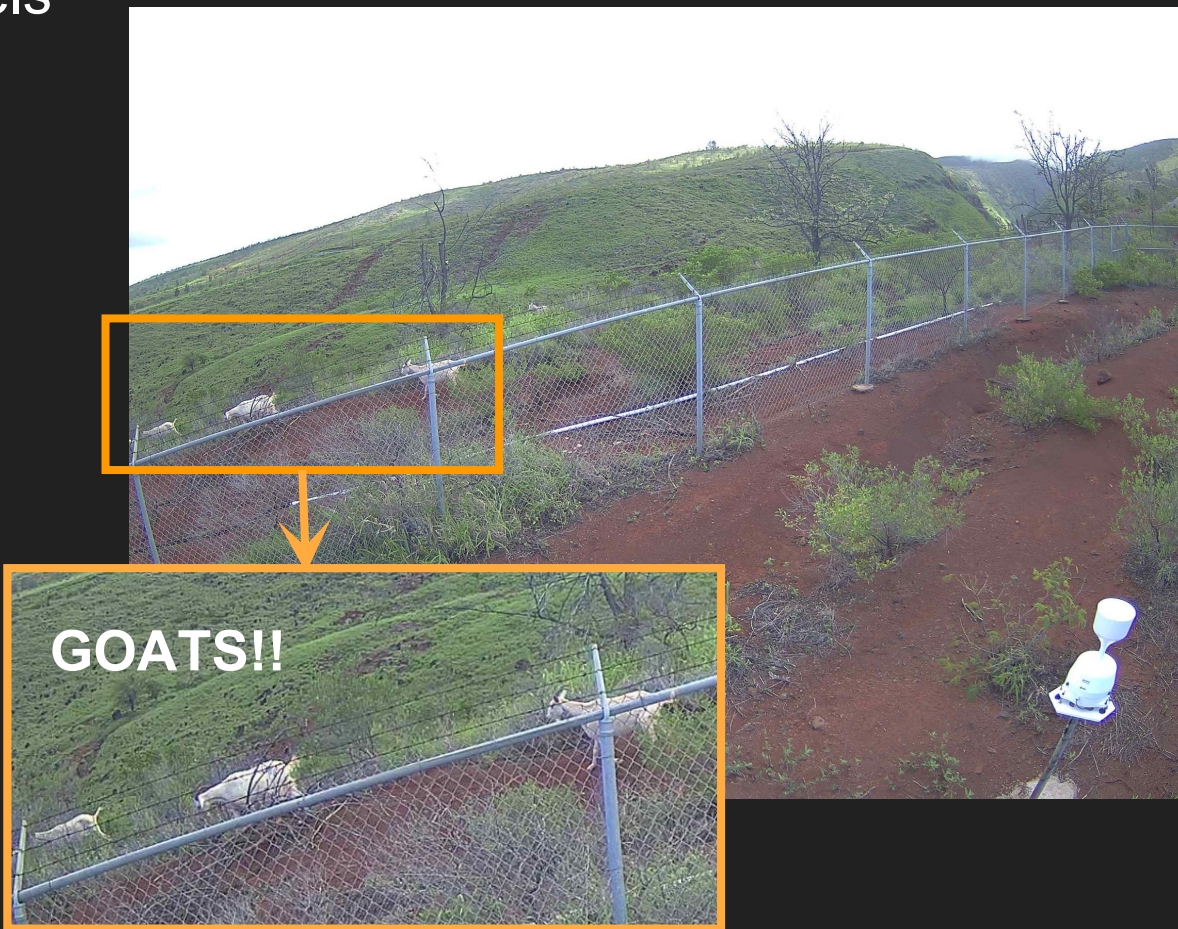


Blue box = discrete volcanic eruptions confirmed by USGS, Red = VLM detected eruption, Grey = VLM detected no eruption

(need a better way to visualize this)

Vision Language Models

- Animal detection with off the shelf VLMs
- Prompt: Is there an animal?
 - Moondream2: (NO)
 - Gpt-4o-mini: (NO)
 - Gpt-4o: (YES)
- Vast majority of the time, it will be false positives.
- This is a rare photo of a true positive
- More work needs to be done, off the shelf general models such as moondream2, gpt-4o-mini, gpt-4o, are insufficient in detecting animals in our environment



Possible Future Directions

- Tentative Hawaii Grande Sites
 - <https://go.hawaii.edu/rbq>
 - https://www.google.com/maps/@20.6951232,-159.0091421,7z/data=!3m1!4b1!4m2!6m1!1s1vthANlcHz-ojSwejtExZ10E6QytVjls?entry=tту&g_ep=EgoyMDI1MDcwNi4wLkXMDSoASAFQAw%3D%3D
 - Focus is on sustainable sites with strong site host/community interest with dynamic viewsheds
- Develop a rigorous way to answer macro-queries
 - e.g. When will the next drought occur
- Investigate Multimodal LLMs
- Look into newer AI/LLM based approaches for anomaly detection
- Anomaly detection for automatic validation of instrument data
- Long term characterization of accessibility and reliability of the test bed
 - Also extend to other instruments
- Test other congestion control methods for data transmission
- Potentially write a paper for PEARC (Practice and Experience in Advanced Research Computing)

Other Commercial Approaches



- <https://www.youtube.com/watch?v=pdb-aZe-1GE>